

qPOTS: A Python Package for Sample-efficient Batch Constrained Multiobjective Bayesian Optimization

Kade E. Carlson¹, Ashwin Renganathan^{1,2}, and Peter E. Bachman¹

¹ Department of Aerospace Engineering, The Pennsylvania State University, United States ² Institute for Computational and Data Sciences, The Pennsylvania State University, United States

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Summary

qPOTS is a Python package for researchers who need to optimize expensive black-box systems with more than one competing objective. In such problems, each experiment, simulation, or hardware test can be costly, and the best answer is rarely a single design but instead a set of trade-off designs that form a Pareto front. Examples include aircraft design, process optimization, materials discovery, and machine learning model selection. qPOTS helps users decide which designs to evaluate next by combining Gaussian-process surrogate models, Thompson sampling, and evolutionary search over sampled Pareto fronts.

The package implements batch Pareto optimal Thompson sampling (qPOTS) for multiobjective Bayesian optimization (Renganathan & Carlson, 2025; Thompson, 1933). It provides tools for unconstrained and constrained benchmark functions, Gaussian-process model fitting and prediction, batch candidate selection via a maximin diversity rule, expected hypervolume tracking, and comparison with standard multiobjective Bayesian optimization acquisition functions. The package also includes qPOTS-D0E, a major multitask extension for decoupled oracle evaluations: when objectives or constraints can be measured separately, the software uses a multitask Gaussian process (MTGP) to transfer information across correlated tasks and then selects only the subset of oracles that are most informative for the current candidate (Álvarez et al., 2012; Bonilla et al., 2007).

qPOTS is released under the GNU General Public License v3.0 and is built on widely used scientific Python libraries including PyTorch, GPyTorch, BoTorch, and pymoo (Balandat et al., 2020; Blank & Deb, 2020; Gardner et al., 2018; Paszke et al., 2019).

Statement of need

Multiobjective optimization is central to scientific and engineering design because improving one objective often degrades another. Classical evolutionary algorithms such as NSGA-II can approximate Pareto fronts accurately but require many objective evaluations (Deb et al., 2002). That cost is prohibitive when each evaluation requires a high-fidelity computational simulation, a physical experiment, or a hardware test that may take hours or days. Bayesian optimization addresses this by fitting probabilistic surrogate models to existing data and using them to select the next most informative experiments (Jones et al., 1998; Rasmussen & Williams, 2006; Shahriari et al., 2016).

Multiobjective Bayesian optimization is substantially harder than its single-objective counterpart. Acquisition functions must reason about a set-valued Pareto frontier rather than a scalar optimum, making them nonconvex, nondifferentiable, or expensive to optimize in batch settings. qPOTS addresses this by replacing direct acquisition-function maximization with posterior-sample optimization: it draws sample paths from the current surrogate model, solves

a comparatively inexpensive multiobjective problem on those samples using evolutionary search, and selects a diverse batch of candidates from the resulting sampled Pareto set (Renganathan & Carlson, 2025). This avoids the need for differentiable analytical acquisitions and naturally yields batches without additional approximation.

A further opportunity arises when the objectives and constraints of a problem can be observed independently. In aerodynamic design, for instance, drag and lift coefficients may be evaluated at multiple operating conditions that require separate solver runs; in chemical processes, different properties may require different laboratory analyses. Existing decoupled Bayesian optimization methods typically assume independent objective models or rely on known scalar oracle costs. qPOTS-D0E (qPOTS with decoupled oracle evaluations) fills this gap by fitting a single multitask Gaussian process over all objectives and constraints jointly, using inter-task posterior correlation to decide whether decoupled evaluations are beneficial, and selecting the most informative oracle subset via a Gaussian mutual-information criterion.

The target audience is researchers and engineers who need reproducible, sample-efficient multiobjective optimization workflows in Python. qPOTS is designed for method developers benchmarking against standard acquisition strategies, and for application researchers optimizing expensive custom black-box functions under limited evaluation budgets, including scenarios where not all objectives need to be queried at every iteration.

State of the field

BoTorch provides a general platform for Bayesian optimization and implements strong multiobjective acquisitions such as qEHVI, qNEHVI, and their logarithmic variants (Ament et al., 2023; Balandat et al., 2020; Daulton et al., 2020). Other influential methods include entropy-search approaches (PESMO, MESMO, JESMO) and lookahead methods such as the hypervolume knowledge gradient (Belakaria et al., 2019; Daulton et al., 2023; Garrido-Merchán & Hernández-Lobato, 2019; Tu et al., 2022). These are important baselines, and several are exposed in qPOTS workflows for direct comparison.

qPOTS occupies a different niche. Rather than contributing a single acquisition function to an existing library, it packages the full research workflow around Pareto optimal Thompson sampling: benchmark functions, constrained and unconstrained examples, batch candidate selection, hypervolume evaluation, baseline comparisons, tests, documentation, and reproducible scripts. A focused package was preferable to a narrow BoTorch primitive because the scholarly contribution spans the sample path Pareto optimization workflow, the maximin diversity-based batch selection rule, the benchmark suite, and the newer decoupled multitask workflow. At the same time, qPOTS deliberately builds on BoTorch and GPyTorch rather than reimplementing Gaussian-process infrastructure.

The qPOTS-D0E extension addresses a specific gap in decoupled multiobjective optimization. Existing decoupled approaches are often cost-weighted and frequently assume independent objective or constraint models. qPOTS-D0E instead uses a multitask Gaussian process with an intrinsic coregionalization kernel, a total-correlation gate $TC(\mathbf{x}) = -\frac{1}{2} \log |\mathbf{R}(\mathbf{x})|$ computed from the posterior correlation matrix $\mathbf{R}(\mathbf{x})$, and a Gaussian mutual-information subset rule to decide which oracles to evaluate at a candidate point. This enables meaningful decoupling even when scalar oracle costs are uniform, provided the tasks exhibit useful posterior correlation.

Software design

The package is organized into four main components. The Function interface wraps built-in benchmarks and user-defined objectives behind a common API. ModelObject manages Gaussian-process fitting and prediction, supporting both independent per-objective GPs (the original qPOTS mode) and a single joint multitask GP for qPOTS-D0E. Acquisition

88 contains the candidate-generation logic for qPOTS and all comparison methods. Utility modules
89 compute hypervolume indicators, filter candidates by diversity and dominance, manage data
90 transformations, and support experiment scripts.

91 This design makes several trade-offs explicit. First, qPOTS uses posterior-sample optimization
92 rather than closed-form acquisition maximization. This introduces stochastic evolutionary-
93 optimization overhead per iteration, but avoids the need for differentiable analytical acquisitions
94 and naturally supports batch candidate generation without additional approximations. Second,
95 the package delegates multiobjective search over sampled posterior paths to NSGA-II via
96 pymoo, keeping the Python interface simple while using a mature and well-tested evolutionary
97 optimizer. Third, precision and device handling are centralized in `config.py`, so users can run
98 CPU-only examples or move tensor computations to CUDA hardware with a single setting
99 change.

100 The qPOTS-D0E multitask extension adds a two-stage decision layer on top of the original
101 candidate selection. In the standard coupled mode, all objectives and constraints are observed
102 at every chosen design. In decoupled mode, the MTGP posterior correlation matrix is used
103 to compute a scalar total-correlation score at the proposed candidate; if this score exceeds
104 a threshold τ , the algorithm selects only an informative subset of oracles by maximizing the
105 Gaussian mutual information $I(\mathbf{Y}_{\mathcal{S}}(\mathbf{x}); \mathbf{Y}_{\bar{\mathcal{S}}}(\mathbf{x}) \mid \mathcal{D})$ between selected and unselected tasks.
106 The MTGP then uses posterior mean imputation to fill unobserved task values and continues
107 the optimization loop. This architecture preserves the original qPOTS sample-path candidate
108 generation while adding principled oracle selection.

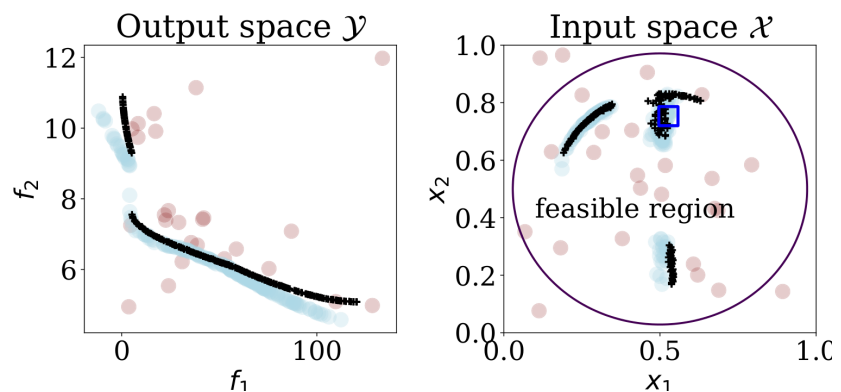


Figure 1: Illustration of qPOTS on a constrained two-objective problem. Shown are the initial training data, the Pareto front approximation, and the batch of candidates proposed by the acquisition strategy.

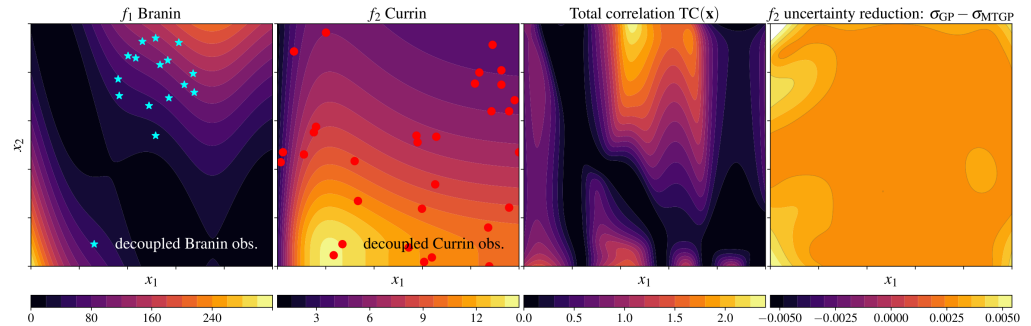


Figure 2: The qPOTS-D0E multitask update uses posterior task correlation to decide when decoupled evaluations are informative. The Branin-Currin example illustrates task-specific observations, total-correlation evolution, and uncertainty reduction from multitask information sharing.

Research impact statement

qPOTS is the reference implementation for the AISTATS 2025 paper introducing Pareto optimal Thompson sampling for batch multiobjective Bayesian optimization (Renganathan & Carlson, 2025). The method was evaluated on eight synthetic and real-world benchmarks against qEHVI, qNEHVI, qLogNEHVI, ParEGO, PESMO, MESMO, JESMO, TS-EMO, and Sobol baselines, demonstrating competitive or superior hypervolume improvements with substantially lower computational cost per iteration compared to Monte Carlo acquisition methods. The package has also supported an applied aerodynamic design study of the NASA Common Research Model, where qPOTS was deployed for 24-dimensional, two-objective aerodynamic design optimization and released with open-source reproducibility materials (Carlson & Renganathan, 2026).

The qPOTS-D0E extension further expands the implementation to correlated objective and constraint tasks with partial observations. In benchmark experiments, qPOTS-D0E is compared against coupled qPOTS, qLogNEHVI, HVKG, PESMO, MESMO, JESMO, and Sobol baselines. The method achieves the highest hypervolume on six of eight synthetic benchmarks and on all four real-world benchmarks tested, while reducing the total number of scalar oracle evaluations by approximately 30% on constrained problems (e.g., 692 versus 976 evaluations on the OSY benchmark) without sacrificing Pareto front quality. The repository provides tests, worked examples, hosted documentation at ReadTheDocs, and PyPI and source installation paths for users who want to reproduce the published workflows or adapt them to new expensive black-box optimization problems.

The broader research community has also begun to use qPOTS as a reference method and benchmark baseline. Recent work has compared against or built on Pareto optimal Thompson sampling in applications including generative molecular design, sustainable process systems, automated class-imbalance learning, diffusion-based Pareto-front refinement, and many-objective Bayesian optimization (Hoteigni & Peitz, 2025; Jiang et al., 2026; Kudva et al., 2026; Muthyala et al., 2025; Wang et al., 2025). These independent uses indicate that the code is serving not only as a reproducibility artifact for the original paper, but also as shared infrastructure for evaluating new multiobjective Bayesian optimization methods.

AI usage disclosure

Anthropic Claude was used to assist with refactoring the code and structuring the Readme.md before submission. Generative AI was not used to design the qPOTS or qPOTS-D0E algorithms, to implement the software, or to generate any benchmark results. The authors are responsible for all technical claims in this manuscript; claims should be verified against the source code,

hosted documentation, the AISTATS qPOTS paper (Renganathan & Carlson, 2025), and the qPOTS-DOE manuscript (soon to be posted on arXiv).

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References

- Álvarez, M. A., Rosasco, L., & Lawrence, N. D. (2012). Kernels for vector-valued functions: A review. *Foundations and Trends in Machine Learning*, 4(3), 195–266. <https://doi.org/10.1561/22000000036>
- Ament, S., Daulton, S., Eriksson, D., Balandat, M., & Bakshy, E. (2023). Unexpected improvements to expected improvement for bayesian optimization. *Advances in Neural Information Processing Systems*, 36, 20577–20612. <https://doi.org/10.52202/075280-0904>
- Balandat, M., Karrer, B., Jiang, D., Daulton, S., Letham, B., Wilson, A. G., & Bakshy, E. (2020). BoTorch: A framework for efficient monte-carlo bayesian optimization. *Advances in Neural Information Processing Systems*, 33, 21524–21538.
- Belakaria, S., Deshwal, A., & Doppa, J. R. (2019). Max-value entropy search for multi-objective bayesian optimization. *Advances in Neural Information Processing Systems*, 32.
- Blank, J., & Deb, K. (2020). pymoo: Multi-objective optimization in python. *IEEE Access*, 8, 89497–89509. <https://doi.org/10.1109/ACCESS.2020.2990567>
- Bonilla, E. V., Chai, K. M. A., & Williams, C. K. I. (2007). Multi-task gaussian process prediction. *Advances in Neural Information Processing Systems*, 20.
- Carlson, K., & Renganathan, A. (2026). Multiobjective aerodynamic design optimization of the NASA common research model. *Aerospace Science and Technology*, 168, 111120. <https://doi.org/10.1016/j.ast.2025.111120>
- Daulton, S., Balandat, M., & Bakshy, E. (2020). Differentiable expected hypervolume improvement for parallel multi-objective bayesian optimization. *Advances in Neural Information Processing Systems*, 33, 9851–9864.
- Daulton, S., Balandat, M., & Bakshy, E. (2023). Hypervolume knowledge gradient: A lookahead approach for multi-objective bayesian optimization with partial information. *Proceedings of the 40th International Conference on Machine Learning*, 202, 7167–7204.
- Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2), 182–197. <https://doi.org/10.1109/4235.996017>
- Gardner, J., Pleiss, G., Weinberger, K. Q., Bindel, D., & Wilson, A. G. (2018). GPyTorch: Blackbox matrix-matrix gaussian process inference with GPU acceleration. *Advances in Neural Information Processing Systems*, 31, 7576–7586.
- Garrido-Merchán, E. C., & Hernández-Lobato, D. (2019). Predictive entropy search for multi-objective bayesian optimization with constraints. *Neurocomputing*, 361, 50–68. <https://doi.org/10.1016/j.neucom.2019.05.025>
- Hotegni, S. S., & Peitz, S. (2025). SPREAD: Sampling-based pareto front refinement via

- 185 efficient adaptive diffusion. *arXiv Preprint arXiv:2509.21058*.
- 186 Jiang, C., Huang, J., & Li, M. (2026). Do we really need to approach the entire pareto front
187 in many-objective bayesian optimisation? *arXiv Preprint arXiv:2604.09417*.
- 188 Jones, D. R., Schonlau, M., & Welch, W. J. (1998). Efficient global optimization of expensive
189 black-box functions. *Journal of Global Optimization*, 13(4), 455–492. <https://doi.org/10.1023/A:1008306431147>
- 190
- 191 Kudva, A., Tang, W.-T., & Paulson, J. A. (2026). Multi-objective bayesian optimization for
192 networked black-box systems: A path to greener profits and smarter designs. *Optimization
193 of Sustainable Process Systems: Multiscale Models and Uncertainties*, 63–89. <https://doi.org/10.1002/9781394205646.ch3>
- 194
- 195 Muthyala, M. R., Sorourifar, F., Tan, T., Peng, Y., & Paulson, J. A. (2025). Generative
196 multiobjective bayesian optimization with scalable batch evaluations for sample-efficient
197 de novo molecular design. *Industrial & Engineering Chemistry Research*, 65(1), 628–642.
198 <https://doi.org/10.1021/acs.iecr.5c03166.s001>
- 199 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z.,
200 Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M.,
201 Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., ... Chintala, S. (2019). PyTorch: An
202 imperative style, high-performance deep learning library. *Advances in Neural Information
203 Processing Systems*, 32, 8024–8035.
- 204 Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT
205 Press.
- 206 Renganathan, A., & Carlson, K. (2025). qPOTS: Efficient batch multiobjective bayesian
207 optimization via pareto optimal thompson sampling. In Y. Li, S. Mandt, S. Agrawal, &
208 E. Khan (Eds.), *Proceedings of the 28th international conference on artificial intelligence
209 and statistics* (Vol. 258, pp. 4051–4059). PMLR. [https://proceedings.mlr.press/v258/
210 renganathan25a.html](https://proceedings.mlr.press/v258/renganathan25a.html)
- 211 Shahriari, B., Swersky, K., Wang, Z., Adams, R. P., & Freitas, N. de. (2016). Taking the
212 human out of the loop: A review of bayesian optimization. *Proceedings of the IEEE*,
213 104(1), 148–175. <https://doi.org/10.1109/JPROC.2015.2494218>
- 214 Thompson, W. R. (1933). On the likelihood that one unknown probability exceeds another in
215 view of the evidence of two samples. *Biometrika*, 25(3-4), 285–294. [https://doi.org/10.
216 1093/biomet/25.3-4.285](https://doi.org/10.1093/biomet/25.3-4.285)
- 217 Tu, B., Gandy, A., Kantas, N., & Shafei, B. (2022). Joint entropy search for multi-objective
218 bayesian optimization. *Advances in Neural Information Processing Systems*, 35, 9922–9938.
- 219 Wang, Z., Wang, S., Ernst, D., & Xiao, C. (2025). Automated class imbalance learning via
220 few-shot multi-objective bayesian optimization with deep kernel gaussian processes. *IEEE
221 Access*. <https://doi.org/10.1109/access.2025.3591034>